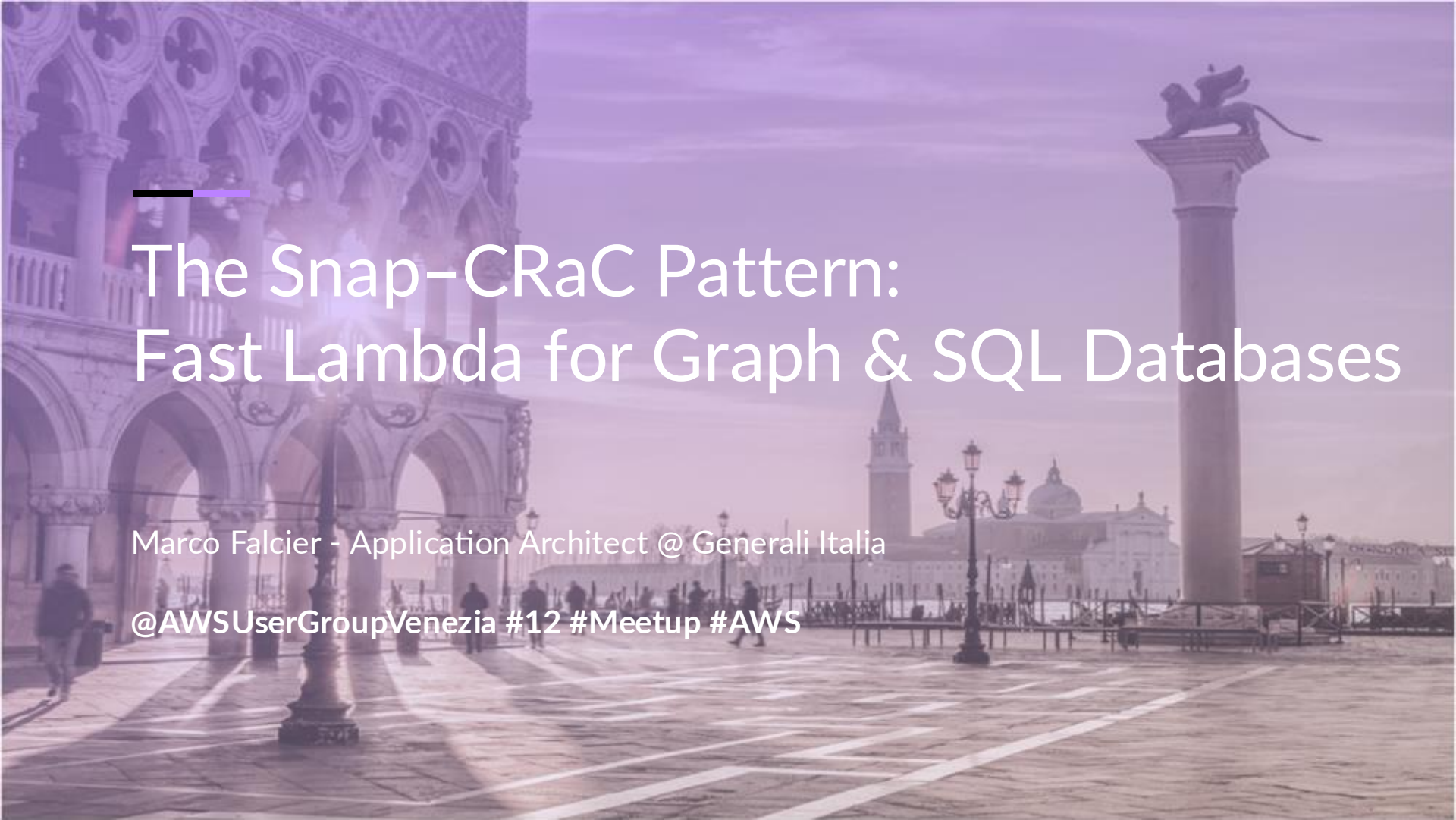

The Snap–CRaC Pattern: Fast Lambda for Graph & SQL Databases

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@AWSUserGroupVenezia #12 #Meetup #AWS

Who am I?

Application Architect @ Generali

- Serverless Enthusiast
- Kotlin Lover
- Neo4j Certified Professional

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In This Talk...

We'll explore a practical pattern to build fast AWS Lambda functions that safely interact with relational and graph databases, using SnapStart and CRaC to control initialization and lifecycle.

The Problem: Cold Starts and Databases

Lambda is fast… until it touches a database

- Cold start = JVM Boot + Optional Inits + DB Setup
- Relational & Graph DBs amplify the issue

Key Message: DB initialization dominates cold start time

Traditional Mitigations (and Limits)

- Provisioned Concurrency
- Smaller dependencies / keep-alive hacks

Limits: cost, partial fixes, slow init

Why Databases Are Special

Not all resources behave the same

- DB connections are:
  - Expensive to create
  - Stateful
  - Often pooled
- Graph DB drivers often perform heavy startup work

Implication: naive Lambda init hurts performance

Enter Snap

What Snapstart Does

- Initializes Lambda once
- Takes a snapshot of memory & execution state
- Restores it on every cold start

Result: JVM cold start ≈ milliseconds

Enter CRaC

CRaC: Coordinated Restore at Checkpoint

- Hooks into JVM lifecycle
- Prepare / restore resources safely

Perfect match for SnapStart

The Snap–CRaC Pattern

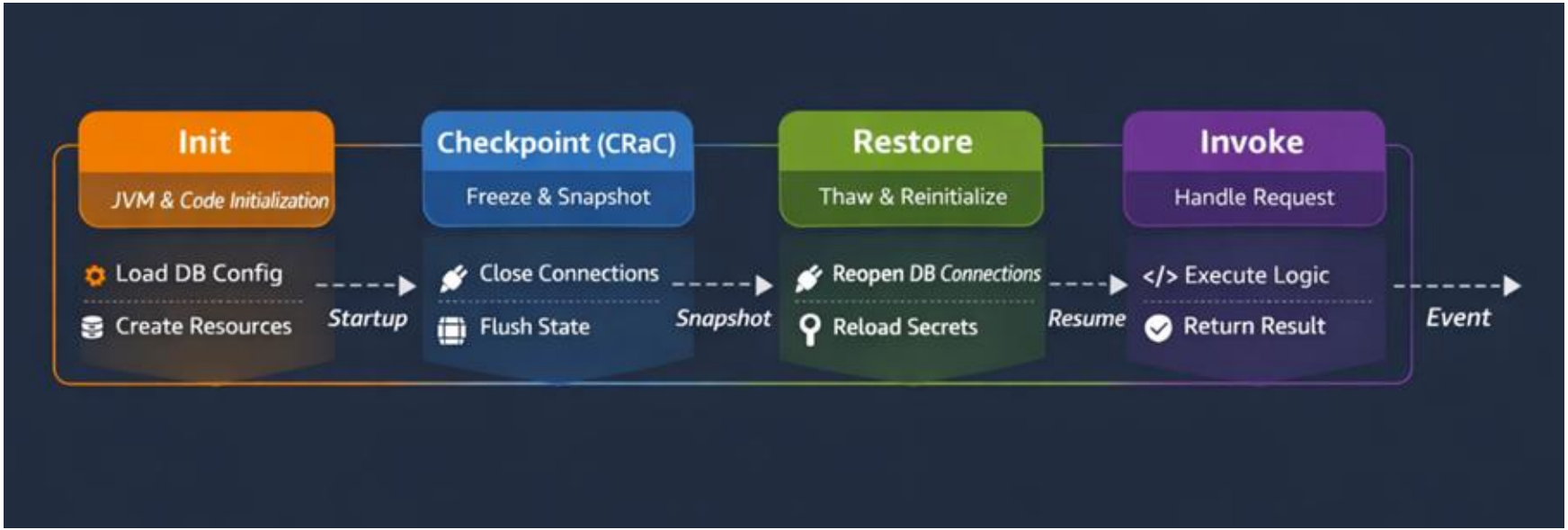
Combining SnapStart + CRaC

Pattern steps:

1. Initialize drivers & config once
2. Close unsafe resources at checkpoint
3. Restore connections on resume

Goal: fast start + safe DB state

Lifecycle View



SQL vs Graph DB Considerations

SQL

- JDBC + connection pools (e.g. HikariCP)
- Pools must be recreated on restore

Graph

- Heavy driver initialization
- Metadata & routing cached once

Why it matters: graph drivers benefits massively from SnapStart

What You Should NOT Snapshot

- Open sockets
- Active transactions
- Thread pools

Rule: snapshot logic, not live connections

Performance Impact

- Cold start: seconds → milliseconds
- Stable latency under burst traffic
- Lower cost vs Provisioned Concurrency

Real impact: predictable p99 latency

When NOT to use This Pattern

- Non-Java runtimes
- Highly dynamic or per-request initialization
- Workloads without cold-start sensitivity

SnapStart ≠ universal solution

CRaC Resource Example

```
public class CRaCResource implements Resource {  no usages

    public CRaCResource () {  no usages
        Core.getGlobalContext().register( r: this);
    }

    private void refreshDriver() {  1 usage
        System.out.println("CRaCResource.refreshDriver()");
    }

    private void closeDriver() {  1 usage
        System.out.println("CRaCResource.closeDriver()");
    }

    public void executeQuery() {  no usages
        System.out.println("CRaCResource.executeQuery()");
    }

    @Override public void beforeCheckpoint(Context<? extends Resource> c) {  no usages
        closeDriver();
    }

    @Override public void afterRestore(Context<? extends Resource> c) {  no usages
        refreshDriver();
    }

}
```

Example in CloudWatch

Neptune Resource Example

```java
@Getter 3 usages
public class NeptuneResource implements Resource  {

    private Session session;

    public NeptuneResource() { 1 usage
        Core.getGlobalContext().register( r: this);
        this.session = NeptuneConfig.driver().session();
        System.out.println("NeptuneResource costruttore ha creato");
    }

    public void executeQuery() { 1 usage
        // implement some logic here
    }

    @Override no usages
    public void beforeCheckpoint(Context<? extends Resource> context) throws Exception {
        System.out.println("Before checkpoint: closing NeptunResource.session");
        this.session.close();
    }

    @Override no usages
    public void afterRestore(Context<? extends Resource> context) throws Exception {
        System.out.println("After restore: refreshing NeptunResource.session");
        this.session = NeptuneConfig.driver().session();
    }
}
```

Key Takeways

- SnapStart removes JVM startup cost
- CRaC makes DB access snapshot-safe
- Works for both SQL & Graph DBs
- Enables truly fast serverless architecture

Closing

Cold starts aren’t the problem,

uncontrolled initialization is.


Questions ?
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